Erin Walshaw

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Education

University of Waterloo September 2024 – August 2029

Bachelor of Computer Science - Artificial Intelligence Specialization, Economics Minor *Waterloo, ON*

Coursework: Algorithm Design, Data Structures, Software Development, Object-Oriented Programming Year: 3A

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, Go, SQL

Frameworks/Tools: React, Next.js, Node.js, Streamlit, PyTorch, scikit-learn, statsmodels, pandas, NumPy

Experience

Manulife | **Business Technology Analyst Intern** September – December 2026

- Built an automatically optimized time-series forecasting system supporting 8+ statistical models, reducing forecasting error by 32.28% by implementing rolling-origin backtests scored with MAE, RMSE, MAPE, AIC/BIC, seasonality, autocorrelation, and trend strength.
- Designed a Streamlit + Plotly interface enabling real-time interactive model comparison with automated parameter optimization built using pandas, NumPy, scikit-learn, statsmodels, and pmdarima.

Poket | **Software Developer Intern** January – April 2026

- Enabled engineering-driven product decisions by deploying GTM/GA4/Microsoft Clarity with 30+ custom event schemas across the Poket Portal.
- Modernized the codebase by upgrading Angular 16 to 17, resolving breaking API changes and dependency conflicts to improve long-term maintainability.
- Shipped production fixes in JavaScript/HTML/CSS that improved UI reliability and resolved functional defects.
- Redesigned the client-facing website using Framer to improve SEO and responsiveness.

University of Waterloo | **Machine Learning Research Assistant** September – December 2025

- Collaborated on exploratory research toward a "constitutional simplex" approach for LLM alignment, analyzing possible geometric latent-space constraints related to honest, harmless, and helpful model behavior.
- Implemented causal intervention pipelines in PyTorch to steer hidden-state vectors toward target behavioral profiles; benchmarked parameter-efficient fine-tuned models against the HHH dataset.

University of Waterloo Futures Lab | **AI + UX Prototyping** September – December 2025

- Built an AI-powered learning tool for hardware design courses using Gemini + Google AI Studio in an interdisciplinary team with graduate researchers.
- Improved prototype usability across 5+ iteration cycles through user-centered design, refining feedback mechanisms and interaction flows.
- Presented architecture and user-testing insights at the Google-UWaterloo Symposium; featured as a student panelist on Google's official blog.

Projects

TinyIR | *Python, scikit-learn* GitHub

- Built a command-line IR engine using tokenization, inverted indexes, and TF-IDF vectorization to rank local text corpora via cosine similarity.
- Implemented persistent index formats and a pytest suite covering tokenization, postings, TF-IDF shape, and ranking; integrated CI via GitHub Actions.

pyglance | *Python (AST, dependency-free)* GitHub

- Built a dependency-free static analyzer using Python's AST to detect unused imports, long functions, unreachable code, and circular imports.
- Implemented recursive directory analysis with import-graph traversal and CI-friendly non-zero exit codes.

Carbon Calories | *Python, React, Next.js, MongoDB, Gemini API* GitHub

- Built a full-stack food-tracking app delivering per-user carbon insights and meal recommendations, integrating a Next.js/React frontend with a Python backend and Gemini API for AI-powered meal analysis.

Leadership

Women in Computer Science Committee | **Director of Finance** May – August 2026

- Managed committee budgets, check requests, and funding for a 100+ member student org.